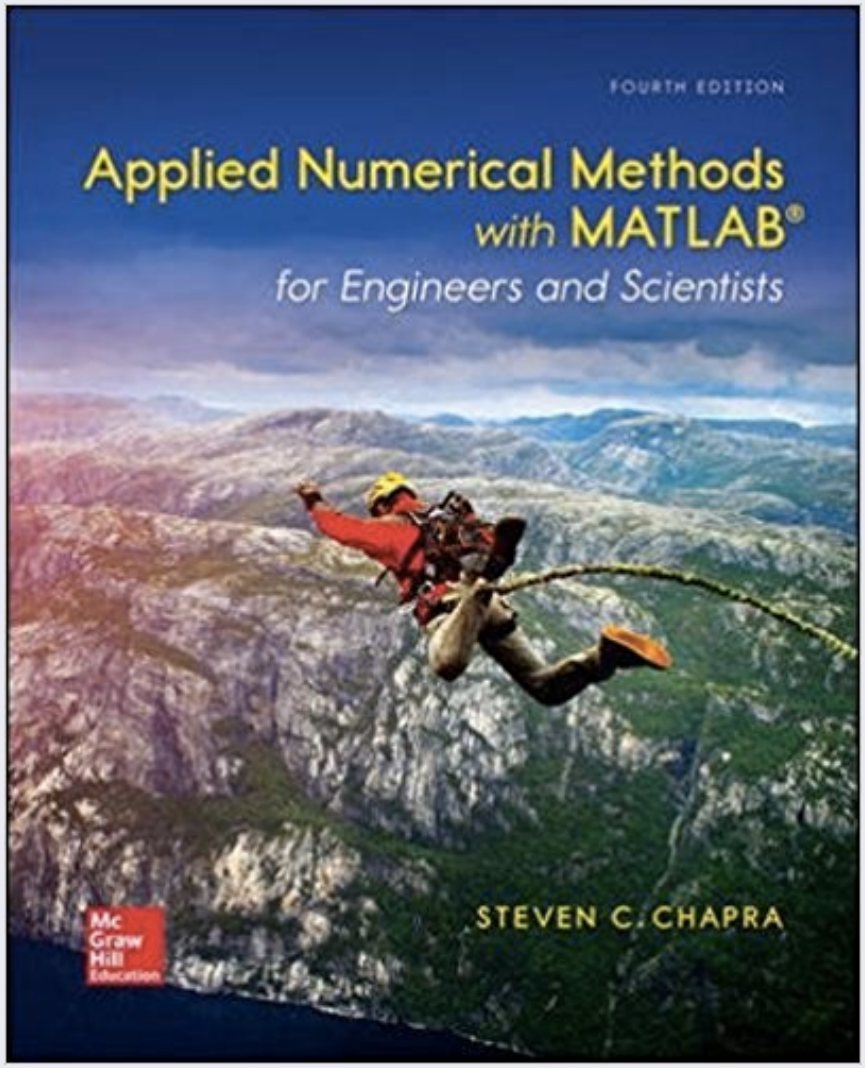
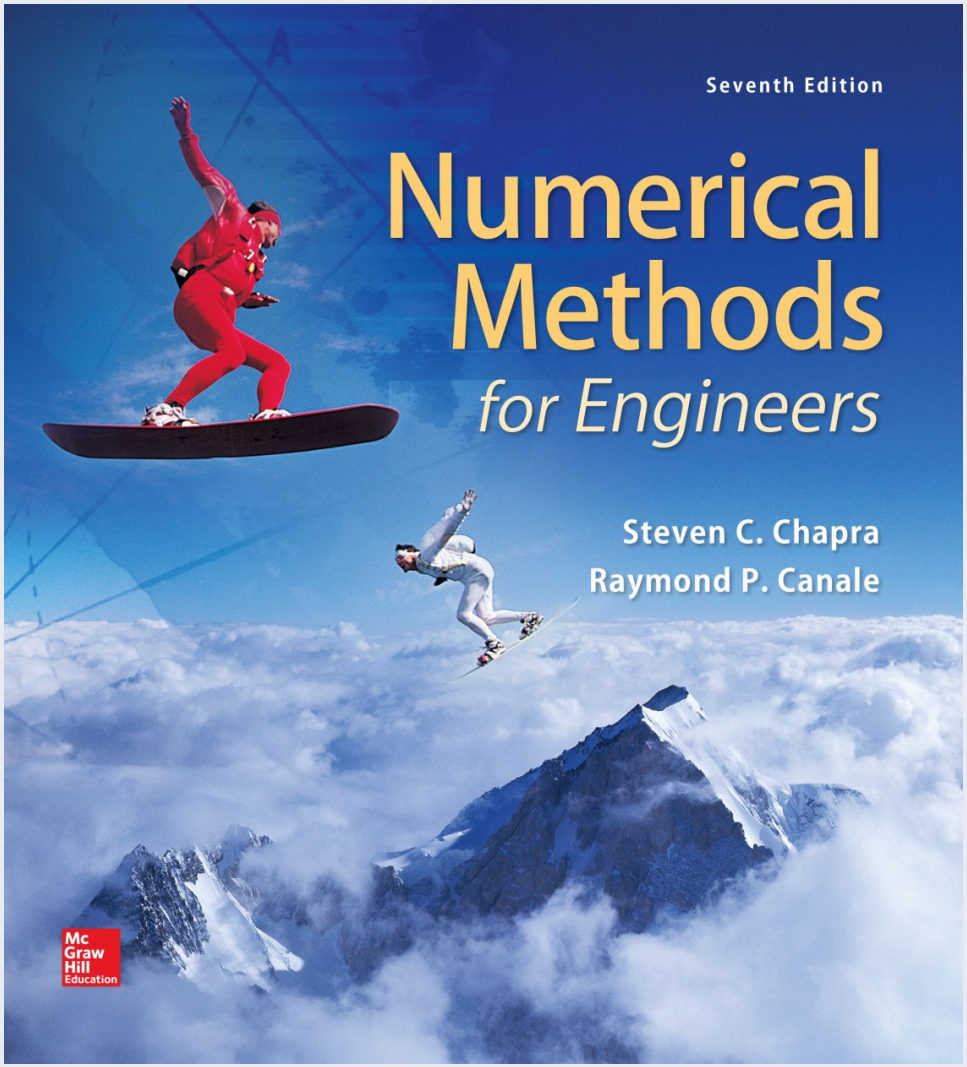
# MATH2059
# Numerical Methods

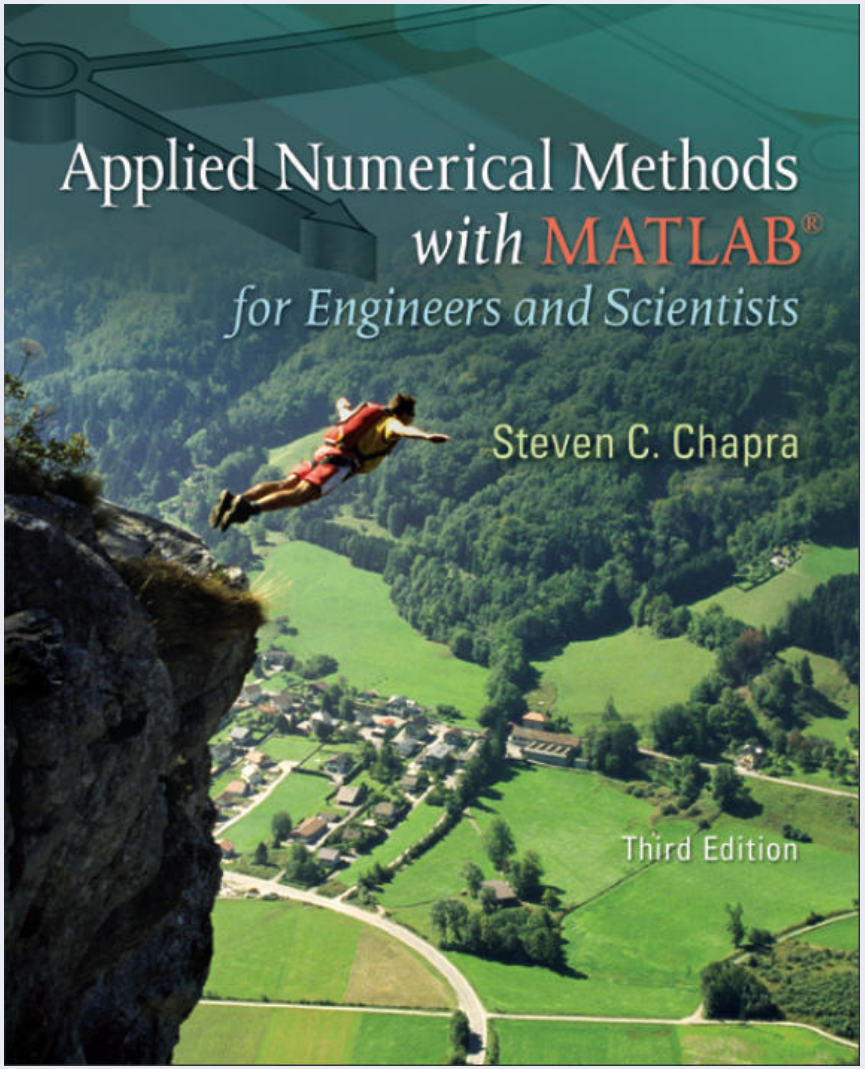
Prof. Dr. Çiğdem Eroğlu Erdem
cigdem.erdem@marmara.edu.tr

# Textbooks


FOURTH EDITION
Applied Numerical Methods
with MATLAB®
for Engineers and Scientists
STEVEN C. CHAPRA
Mc Graw Hill Education


Seventh Edition
Numerical
Methods
for Engineers
Steven C. Chapra
Raymond P. Canale
Mc Graw Hill Education

# Older Editions



# Course Web Page

- https://classroom.google.com
- Class code: o76m5g6
- The Syllabus has been uploaded!
- Visit the class web regularly page to follow:
  - Lecture slides
  - Assignments
  - Announcements

# Course Assistant

- Serap Korkmaz
  - E-mail: serap.korkmaz@marmara.edu.tr

# Course Objectives

- Introduce:
  - Basic numerical methods and
  - Their applications in engineering

# Learning Outcomes

- Learning Objective 1:
  - Use the MATLAB programming language and toolboxes to implement numerical algorithms.
- Learning Objective 2:
  - Solve nonlinear equations with a single unknown using numerical methods.
- Learning Objective 3:
  - Solve systems of linear equations using numerical methods.

# Learning Outcomes

- Learning Objective 4:
  - Apply basic principles of optimization using numerical methods.
- Learning Objective 5:
  - Apply interpolation and regression to fit a curve to data obtained in engineering applications.
- Learning Objective 6:
  - Apply numerical methods for differentiation, integration and differential equations.

# Grading

- Midterm Exam 30%
- Quizzes 20%
- Attendance-Quizzes (Polls) 10%
- Final Exam 40%

- Homeworks will be given for self-study, which will not be graded.

# Quizzes

- There will be at least 4 announced quizzes.
- There is no make-up for quizzes.

# Homeworks

- At least three homeworks will be assigned that require MATLAB programming.
- Homeworks will be given for self-study, which will not be graded.
- Install MATLAB to your PC using your Marmara University e-mail address.

# MATLAB

- Web Page: www.mathworks.com
- You can install to your PC:
  - Create an account using your Marmara University e-mail address.
  - Install MATLAB
  - You can also use MATLAB online.

# MATLAB

- Tutorials
  - Open an account at: www.mathworks.com
  - Go to MATLAB Academy: https://matlabacademy.mathworks.com/
  - Enroll in the self-paced interactive course (takes 2 hours to complete):
    - MATLAB Onramp (free course)